

KRITAGYA TIMSINA

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PROFESSIONAL SUMMARY

Purpose-driven Computer Engineering student with strong interest in full-stack web development, AI-powered applications, and scalable software systems. Experienced in building modern web applications using **Next.js, React, TypeScript, Node.js, Express.js, MongoDB, PostgreSQL, and Python**. Proven ability to design responsive user-focused products, integrate authentication and payment systems, and develop AI-assisted applications with real-time functionality. Passionate about leveraging software and machine learning to solve practical problems through clean architecture, efficient backend systems, and intuitive user experiences.

EDUCATION

Khwopa College of Engineering Libali Bhaktapur, Nepal
Bachelor in Computer Engineering (BCT), **Tribhuvan University Affiliated** Expected Graduation: May 2026

PROJECTS

AI Dental Assistant App (github.com/kritagyatimsina023/dentelligent.git) Personal Project

- Built a full-stack AI-powered dental assistance platform where users can interact with a voice AI agent for dental-related queries and support.
- Implemented appointment booking, doctor availability management, conflict handling, and real-time appointment status updates.
- Integrated secure authentication and user management with Clerk and connected payment workflow for appointment handling.
- Designed scalable frontend and backend architecture using Next.js, TanStack Query, Zustand, Prisma ORM, PostgreSQL, Clerk, and VAPI.

Emotion-Based Symbolic Music Generation (github.com/kritagyatimsina023/aiMusicBackend.git)

Major project

- Developed an AI-driven music generation system where users provide lyrics and captions, and the system generates music based on detected emotional context.
- Applied emotion understanding through the Valence-Arousal model to classify lyrical mood and align generated music with user-provided captions.
- Implemented a 3-stage generation pipeline allowing users to generate multiple music variations including Stage-1, Stage-2, and GAN-based Stage-3 outputs.
- Built the full-stack application using React.js, TypeScript, Firebase Auth, Node.js, Express.js, MongoDB, Flask, and Python-based ML routes.

Real-Time Chat Application (github.com/kritagyatimsina023/chatter.git) Personal Project

- Developed a real-time messaging platform allowing users to chat instantly with selected users and retrieve complete chat history dynamically.
- Implemented JWT-based authentication and secure user session handling.
- Integrated Socket.io for real-time message delivery and notification handling.
- Built using React.js, Zustand, Axios, Node.js, Express.js, MongoDB, and Socket.io.

Query AI (github.com/kritagyatimsina023/nextGPT.git) Personal Project

- Built an AI-powered conversational web application that generates responses to user queries in a ChatGPT-like interface.
- Integrated Gemini 2.5 Flash API for real-time AI response generation.
- Added regional and global payment support using **eSewa** and Stripe.
- Developed with Next.js, MongoDB, AI API integration, and payment gateway workflows.

Spylt Milk Clone Website (github.com/kritagyatimsina023/spylt-milk.git) Frontend Project

- Designed and developed a highly interactive and responsive clone website focused on modern product presentation and animation-driven user experience.
- Implemented advanced scroll-based and text-based animations using GSAP ScrollTrigger and SplitText.
- Built using Next.js, GSAP, and responsive frontend design principles.

MovieMate (github.com/kritagyatimsina023/movie-mate.git)

Frontend Project

- Developed a movie information platform that allows users to search and explore current movie details dynamically.
- Integrated external movie data through the TMDB API and implemented responsive UI behavior.
- Built using React.js, Zustand, and TMDB API integration.

Code Book (github.com/kritagyatimsina023/codebook.git)

Personal Project

- Built an e-commerce themed online book purchasing application focused on core shopping functionality and user experience.
- Implemented JWT-based authentication, protected routing, and state management for product and cart handling.
- Developed using React.js, Redux, React Router DOM, JSON Server, and frontend authentication workflows.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS, C/C++

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, PyTorch, Pandas, NumPy, Zustand, TanStack Query, Redux

Databases & ORM: MongoDB, PostgreSQL, MySQL, Prisma ORM

Tools & Platforms: Git, GitHub, Firebase Auth, Clerk, Socket.io, RESTful API, VAPI, Stripe, eSewa

Soft Skills: Agile Development